More Projects

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1. Co-Creating with GPT-5 – By Lablab.ai

Project: Eco Mentor ([View Here])

Role: Full-Stack Developer

- Developed an **GPT-5-powered sustainability mentor** for kids with interactive chat, quizzes, and real-life eco challenges.
- Integrated **gamification features** including points, badges, and progress tracking.
- **Tech Stack:** FastAPI, React, GPT-5 API, Railway, Vercel.

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2. HackHub 2025 – By SparkHub

Project: Exam Companion – Exam Preparation Assistant ([View Here])

Role: Full-Stack Developer

- Built a **smart web application** to help students plan, track, and manage exam study schedules.
- Developed **interactive dashboards**, schedule export options, and a fully **responsive interface** for multiple devices.

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3. Qloo LLM Hackathon – By Qloo

Project: MoodFlow – Mood-Based Recommender ([View Here])

Role: Full-Stack Developer

- Created **MoodFlow**, an **LLM-powered lifestyle recommender** using **Qloo's Taste AI™** and OpenAI APIs.
- Suggested **mood-based recommendations** for books, music, and movies.
- Deployed the solution on **Streamlit** for instant access and user testing.

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4. RAISE Your Hack Hackathon – By Lablab.ai

Project: Legal E-Commerce AI Agent ([View Here])

Role: Retriever Specialist

- Developed an **Urdu AI legal assistant** using **Groq API, LangChain, and Llama models**.
- Implemented **clause extraction, legal Q&A, and notice drafting** pipelines.
- Deployed the application on **Hugging Face** for open access.

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5. NASA Space Apps Challenge 2025 – By NASA

Project: Exoplanet Hunter ([View Here])

Role: Team Lead & Machine Learning Developer

- Built an **ML model using NASA's TESS mission dataset** to identify exoplanet candidates.
- Achieved **89% classification accuracy** using the XGBoost algorithm.
- Designed and deployed a **Streamlit web app** for real-time exoplanet prediction and data visualization.

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6. Polarization Mapper – Hackathon Project

Role: AI Developer

- Created a **web-based analytics tool** to measure and visualize **polarization in online discussions** using natural language processing.
- Designed an **interactive dashboard** for exploring polarization trends across social media data.

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7. AgroVision AI – Plant Disease Detection Application

Role: Computer Vision Researcher

- Developed **AgroVision AI**, a **lightweight YOLO-based plant disease detection system** for agricultural use.
- Trained on **50,000+ annotated plant images** across 25 disease classes.
- Achieved **99% precision using YOLOv12**, ensuring high-speed diagnosis and efficiency in real-world settings.
- Designed for **early disease detection, crop management, and sustainable agriculture**.

8. Automated Diabetic Retinopathy Screening with Swin Transformer

- Built a **Swin Transformer (shifted-window attention)** model to **detect and grade diabetic retinopathy** into **5 severity classes** using the **APTOS 2019 Blindness Detection** retinal fundus dataset.
- Used **transfer learning (pretrained Swin weights)** plus **image preprocessing/augmentation** (denoising, contrast enhancement, geometric/color transforms) to improve robustness and generalization.
- Improved training with **adaptive learning-rate scheduling** and **class-imbalance handling** (**weighted sampling + focal loss**

9. Lung Cancer Detection from CT Scans using Hybrid Deep Learning and Explainable AI

Developed a CT-scan-based lung cancer classification system to distinguish **benign, malignant, and normal cases** using a hybrid deep learning pipeline. The project combines **InceptionV3 transfer learning**, **attention mechanisms**, and handcrafted texture features such as **GLCM and LBP** to improve feature representation. Explainability was added using **Grad-CAM**, allowing visualization of the image regions influencing model predictions. The model achieved strong validation performance with around **98.9% accuracy** and **0.997 ROC-AUC** on the validation set.

- Built a **hybrid lung cancer detection model** using CT scan images by combining **InceptionV3 transfer learning**, attention blocks, and handcrafted texture features including **GLCM and LBP** for multi-class classification.
- Implemented a complete medical imaging pipeline with **image preprocessing, adaptive Gaussian denoising, feature extraction, class balancing, model training, evaluation, and confusion matrix/ROC-AUC analysis**.

- Integrated **Grad-CAM explainable AI** to visualize prediction-relevant regions in CT scans, improving model interpretability for medical AI applications; achieved approximately **98.9% validation accuracy** and **0.997 ROC-AUC**.

10. Heart Sound Classification using Lightweight Deep Learning Models

Developed an end-to-end deep learning pipeline to classify **phonocardiogram heart sound recordings** as **normal or abnormal** using the **PhysioNet/CinC 2016 dataset**. The project compared lightweight architectures, including **Tiny 2D CNN, 1D CNN, and YAMNet-based transfer learning**, with evaluation focused on accuracy, F1-score, inference time, and energy efficiency.

CV Bullet Points

- Built a complete **heart sound classification pipeline** using Python, TensorFlow, Librosa, and Scikit-learn, including audio preprocessing, segmentation, feature extraction, model training, and evaluation.
- Developed and compared **three lightweight deep learning models**: Tiny 2D CNN on log-Mel spectrograms, 1D CNN on raw waveform segments, and YAMNet embeddings with a dense classifier head.
- Achieved **92.44% accuracy** with the Tiny 2D CNN model, while also analyzing **F1-score, recall, inference time, and energy consumption** for lightweight/real-time deployment suitability.

11.EfficientUNetViT-Based Breast Tumor Segmentation

This project focused on developing a deep learning-based system for **automatic breast tumor segmentation from ultrasound images**. The goal was to accurately identify and outline tumor regions in medical scans, which is an important task in computer-aided diagnosis and medical image analysis.

The project followed a complete machine learning workflow, starting from **dataset preparation and preprocessing**. Breast ultrasound images and their corresponding segmentation masks were loaded, resized, normalized, and prepared for model training. Data augmentation techniques were applied to improve model generalization and reduce overfitting, especially because medical imaging datasets are often limited in size.

Multiple deep learning architectures were implemented and compared, including **EfficientUNetViT, DeepLabV3+, MobileNetV2-UNet**, and ensemble-based approaches. These models were trained to learn pixel-level tumor boundaries and generate accurate segmentation masks. The project also included prediction visualization, where original ultrasound images, ground-truth masks, and predicted masks were compared to evaluate model behavior visually.

Model performance was evaluated using standard medical image segmentation metrics such as **Dice score, IoU, precision, recall, specificity, and pixel accuracy**. Among the tested models, **EfficientUNetViT achieved the best performance**, reaching **95.9% pixel accuracy**, **0.718 Dice score**, and **0.647 IoU**, showing strong capability in detecting and localizing tumor regions.

Overall, this project demonstrates practical experience in **medical image segmentation, deep learning model comparison, computer vision, preprocessing, evaluation, and visualization**, with a focus on building AI systems that can support medical image analysis.

12. Acute Mountain Sickness Detection using Hyperdimensional Computing

CV Points

- Developed a lightweight **Hyperdimensional Computing-based binary classification pipeline** to detect Acute Mountain Sickness from physiological and contextual features such as SpO_2, heart rate, CO level, time, event, and subject information.
- Implemented end-to-end preprocessing including **label creation, categorical encoding, missing-value handling, feature scaling, mutual information analysis, and hypervector-based sample representation**.
- Compared multiple HDC configurations using **pseudo-random, Sobol, and Hadamard hypervector generation**, achieving a best accuracy of **81.8%** and outperforming an **RBF-SVM baseline** on the same test setup.

This project focuses on detecting **Acute Mountain Sickness (AMS)** using a lightweight machine learning approach based on **Hyperdimensional Computing (HDC)**. The AMS total score was converted into a binary target, where scores below 2 represented **No AMS** and scores of 2 or higher represented **AMS**.

The pipeline used key physiological and contextual indicators, including oxygen saturation, heart rate, carbon monoxide level, event name, time, and subject ID. The data was cleaned, categorical features were encoded, missing values were handled, and features were normalized before training. A mutual information analysis was also performed to understand the contribution of selected features.

The main model encoded each sample into high-dimensional bipolar hypervectors and classified test samples using cosine similarity with class hypervectors. Different hypervector generation techniques and dimensions were tested to evaluate performance. The best AMS-HD configuration achieved **81.8% accuracy**, with balanced macro precision, recall, and F1-score of around **81.7%**, showing better performance than the SVM baseline.